Lecture 1
Quadratic separation of a Gaussian

A complete density proof of Gaussian conditioning

Probabilistic Machine Learning 2027-1

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Section 1

Block elimination separates the joint quadratic

The joint density sets two proof obligations

Supplied model Prepared live: Gaussian blocks and density

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} = \Sigma^T \succ 0$$

$$X_i, \mu_i \in \mathbb{R}^{d_i}, \quad \Sigma_{ij} \in \mathbb{R}^{d_i \times d_j}, \quad d = d_1 + d_2, \quad d_i \geq 1$$

$$p(x_1, x_2) = \mathcal{N}_d(x; \mu, \Sigma)$$
$$= (2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}} \exp\left(-\tfrac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right)$$

> Separating the quadratic is one obligation. The normalizer must split as well.

Positive definiteness makes the observed block invertible

$$H := \Sigma_{22}, \quad z \in \mathbb{R}^{d_2} \setminus \{0\}$$
$$z^T H z = \begin{pmatrix} 0 \\ z \end{pmatrix}^T \Sigma \begin{pmatrix} 0 \\ z \end{pmatrix} > 0 \quad \Rightarrow \quad H = H^T \succ 0$$
$$B := \Sigma_{12} H^{-1}, \quad S := \Sigma_{11} - B H B^T = S^T$$
$$BH = \Sigma_{12}, \quad H B^T = \Sigma_{21}$$
$$S = \Sigma_{11} - \Sigma_{12} H^{-1} \Sigma_{21}$$

> S is the live residual covariance (the Schur complement of H). Its strict positivity still needs proof here.

The elimination matrix has an explicit inverse

$$T := \begin{pmatrix} I_{d_1} & -B \\ 0 & I_{d_2} \end{pmatrix}, \quad U := \begin{pmatrix} I_{d_1} & B \\ 0 & I_{d_2} \end{pmatrix}$$

$$TU = \begin{pmatrix} I_{d_1} & B-B \\ 0 & I_{d_2} \end{pmatrix} = I_d$$

$$UT = \begin{pmatrix} I_{d_1} & -B+B \\ 0 & I_{d_2} \end{pmatrix} = I_d \quad \Rightarrow \quad T^{-1} = U$$

> Block elimination is reversible; no covariance or independence argument was used.

Left elimination removes the upper covariance block

Carried forward R1.3

$$BH = \Sigma_{12}, \quad HB^T = \Sigma_{21}, \quad S = \Sigma_{11} - BHB^T$$

$$T\Sigma = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} \Sigma_{11} & BH \\ HB^T & H \end{pmatrix}$$

$$= \begin{pmatrix} \Sigma_{11} - BHB^T & BH - BH \\ HB^T & H \end{pmatrix}$$

$$T\Sigma = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}$$

The upper cross block vanishes. This single multiplication is not yet the covariance congruence.

Right elimination completes the covariance congruence

Carried forward R1.5 and transpose of R1.4

$$T\Sigma = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}, \quad T^T = \begin{pmatrix} I & 0 \\ -B^T & I \end{pmatrix}$$

$$\begin{aligned} T\Sigma T^T &= \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}\begin{pmatrix} I & 0 \\ -B^T & I \end{pmatrix} \\ &= \begin{pmatrix} S & 0 \\ HB^T - HB^T & H \end{pmatrix} \\ D := T\Sigma T^T &= \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix} \end{aligned}$$

The two multiplications form a congruence: $T\Sigma T^T$. They do not form a similarity transformation.

Invertible congruence proves the residual covariance positive definite

Carried forward R1.4 and R1.6

$$D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}, \quad T \text{ invertible}, \quad \Sigma \succ 0$$

$$w \neq 0 \quad \Rightarrow \quad T^T w \neq 0$$

$$w^T D w = \left(T^T w\right)^T \Sigma \left(T^T w\right) > 0 \quad \Rightarrow \quad D \succ 0$$

$$a \in \mathbb{R}^{d_1} \setminus \{0\}, \quad w = \begin{pmatrix} a \\ 0 \end{pmatrix}$$

$$a^T S a = w^T D w > 0 \quad \Rightarrow \quad S \succ 0$$

Both S and H have inverses and positive determinants. This follows from strict joint positivity.

Inverting the congruence fixes the order of the factors

Carried forward R1.6; inverses justified in R1.7

$$D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$$

$$D^{-1} = T^{-T}\Sigma^{-1}T^{-1}, \quad T^{-T} := \left(T^{-1}\right)^T$$

$$T^T D^{-1} T = \Sigma^{-1}$$

$$\Sigma^{-1} = T^T \begin{pmatrix} S^{-1} & 0 \\ 0 & H^{-1} \end{pmatrix} T$$

> The factors do not commute. The order follows from undoing the congruence.

Elimination turns the centered vector into a residual and an observation

$$u := x_2 - \mu_2, \quad r := x_1 - \mu_1 - Bu$$
$$T(x-\mu) = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} x_1-\mu_1 \\ u \end{pmatrix} = \begin{pmatrix} x_1-\mu_1-Bu \\ u \end{pmatrix} = \begin{pmatrix} r \\ u \end{pmatrix}$$

r is the centered residual. Its coordinates contain x_2; the original first block is still x_1.

The joint quadratic is the sum of two block quadratics

Carried forward R1.8 and R1.9

$$\Sigma^{-1} = T^T D^{-1} T, \quad T(x-\mu) = \begin{pmatrix} r \\ u \end{pmatrix}$$
$$D^{-1} = \begin{pmatrix} S^{-1} & 0 \\ 0 & H^{-1} \end{pmatrix}$$

$$(x-\mu)^T \Sigma^{-1} (x-\mu) = [T(x-\mu)]^T D^{-1} [T(x-\mu)]$$
$$= \begin{pmatrix} r \\ u \end{pmatrix}^T \begin{pmatrix} S^{-1} & 0 \\ 0 & H^{-1} \end{pmatrix} \begin{pmatrix} r \\ u \end{pmatrix}$$
$$(x-\mu)^T \Sigma^{-1} (x-\mu) = r^T S^{-1} r + u^T H^{-1} u$$

The exponent separates exactly. A normalized density equality has not yet been proved.

Retrospective: elimination separates the quadratic

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}, \quad H \succ 0, \quad S \succ 0$$
$$(x-\mu)^T \Sigma^{-1} (x-\mu) = r^T S^{-1} r + u^T H^{-1} u$$
$$u = x_2 - \mu_2, \quad r = x_1 - \mu_1 - Bu$$

TAKE HOME

Reversible elimination yields valid residual and observed covariance blocks. The inverse congruence separates their quadratic contributions.

Section 2

Normalizers identify the conditional and marginal densities

From separated exponents to probability laws

Unit-determinant elimination splits the covariance determinant

Carried forward R1.4 and R1.6

$$T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}, \quad D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$$

$$\det T = \det I_{d_1} \det I_{d_2} = 1$$

$$\det D = (\det T)(\det \Sigma)(\det T^T) = \det \Sigma$$

$$\det \Sigma = \det S \det H, \quad \det \Sigma, \det S, \det H > 0$$

Elimination changes the covariance blocks while preserving this determinant. Positivity licenses positive square roots.

The dimensions split the remaining Gaussian normalizer

Carried forward R1.2, R1.7 and R2.2

$$d = d_1 + d_2, \quad \det \Sigma = \det S \det H, \quad S, H \succ 0$$

$$\begin{aligned}
&(2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}} \\
&= (2\pi)^{-\frac{d_1+d_2}{2}} (\det S \det H)^{-\frac{1}{2}} \\
&(2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}} \\
&= \left[(2\pi)^{-\frac{d_1}{2}} (\det S)^{-\frac{1}{2}}\right] \left[(2\pi)^{-\frac{d_2}{2}} (\det H)^{-\frac{1}{2}}\right]
\end{aligned}$$

> Each block receives its own determinant factor and its own dimension factor.

The full density equals two normalized Gaussian factors

Carried forward R1.3, R1.7 and R1.9

$$u = x_2 - \mu_2, \quad r = x_1 - (\mu_1 + Bu), \quad S, H \succ 0$$

$$\begin{aligned} g_1(x_1; x_2) &:= (2\pi)^{-\frac{d_1}{2}} (\det S)^{-\frac{1}{2}} \exp\!\left(-\tfrac{1}{2} r^T S^{-1} r\right) \\ &= \mathcal{N}_{d_1}(x_1; \mu_1 + Bu, S) \\ g_2(x_2) &:= (2\pi)^{-\frac{d_2}{2}} (\det H)^{-\frac{1}{2}} \exp\!\left(-\tfrac{1}{2} u^T H^{-1} u\right) \\ &= \mathcal{N}_{d_2}(x_2; \mu_2, H) \\ p(x_1, x_2) &= g_1(x_1; x_2) g_2(x_2) \end{aligned}$$

For fixed x_2, g_1 is a Gaussian density in x_1. The factor names do not yet identify the marginal and conditional.

Integrating the first factor identifies the marginal

Carried forward R2.4

$$p(x_1, x_2) = g_1(x_1; x_2) g_2(x_2), \quad g_1(x_1; x_2) = \mathcal{N}_{d_1}(x_1; \mu_1 + Bu, S)$$

$$S \succ 0, \quad m := \mu_1 + B(x_2 - \mu_2) \ \text{ is fixed}$$

$$\int_{\mathbb{R}^{d_1}} g_1(x_1; x_2)\, \mathrm{d}x_1 = \int_{\mathbb{R}^{d_1}} \mathcal{N}_{d_1}(x_1; m, S)\, \mathrm{d}x_1 = 1$$

$$\begin{aligned} p(x_2) &= \int_{\mathbb{R}^{d_1}} p(x_1, x_2)\, \mathrm{d}x_1 \\ &= g_2(x_2) \int_{\mathbb{R}^{d_1}} g_1(x_1; x_2)\, \mathrm{d}x_1 \\ &= g_2(x_2) = \mathcal{N}_{d_2}(x_2; \mu_2, H) \end{aligned}$$

Normalization identifies g_2 as the marginal; no conditional formula was assumed.

Dividing by the positive marginal identifies the conditional

$$p(x_2) = g_2(x_2) = \mathcal{N}_{d_2}(x_2; \mu_2, H) > 0 \quad \left(x_2 \in \mathbb{R}^{d_2}\right)$$

$$p(x_1|x_2) := \frac{p(x_1, x_2)}{p(x_2)}$$

$$= \frac{g_1(x_1; x_2) g_2(x_2)}{g_2(x_2)} = g_1(x_1; x_2)$$

$$= \mathcal{N}_{d_1}(x_1; \mu_1 + B(x_2 - \mu_2), S)$$

$$p(x_1, x_2) = p(x_1|x_2) p(x_2)$$

The positive-density ratio gives a conditional-density version for every x_2. Its normalization was already proved.

The conditional mean retains the observed-block dependence

Carried forward R2.6

$$p(x_1|x_2) = \mathcal{N}_{d_1}(x_1; \mu_1 + B(x_2 - \mu_2), S)$$

$$\mu_{1|2}(x_2) = \mu_1 + B(x_2 - \mu_2), \quad \Sigma_{1|2} = S$$

$$r = x_1 - \mu_{1|2}(x_2)$$

The mean is affine in x_2 and can depend on the observation; the covariance does not. A separated residual quadratic does not assert independence. This exact density proof assumes a joint Gaussian with $\Sigma \succ 0$; it does not establish adequacy of the supplied model.

Retrospective: normalization completes Gaussian conditioning

$$p(x_1, x_2) = g_1(x_1; x_2) g_2(x_2)$$
$$g_1 = \mathcal{N}_{d_1}(x_1; \mu_1 + B(x_2 - \mu_2), S), \quad g_2 = \mathcal{N}_{d_2}(x_2; \mu_2, H)$$
$$\int_{\mathbb{R}^{d_1}} g_1(x_1; x_2)\, \mathrm{d}x_1 = 1 \quad \Rightarrow \quad p(x_2) = g_2(x_2)$$
$$\frac{p(x_1, x_2)}{p(x_2)} = g_1(x_1; x_2) = p(x_1 | x_2)$$

TAKE HOME The quadratic, determinant and dimension factors all split. Integration then division identify the Gaussian marginal and conditional.